

A

B

C

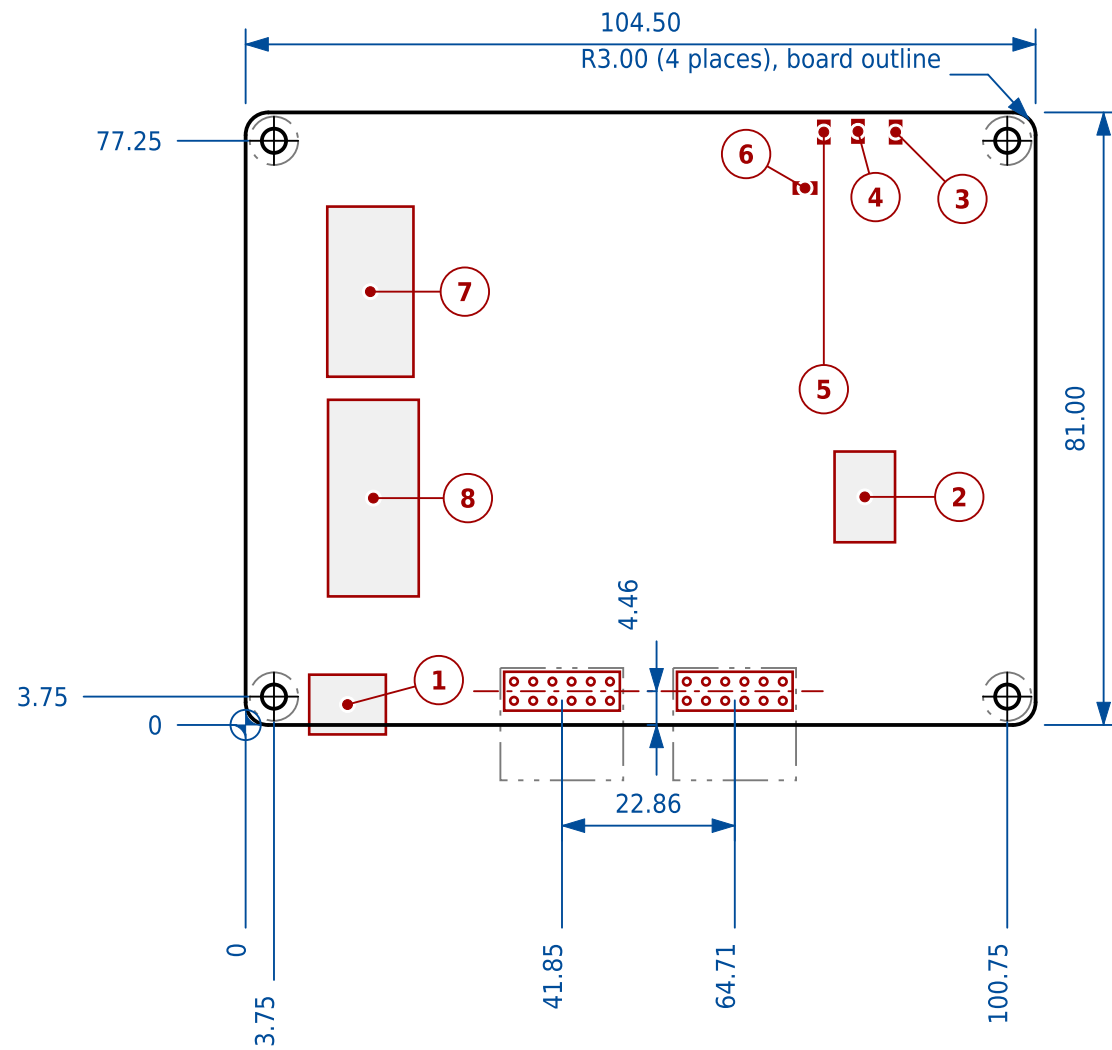
D

A

B

C

D



NOTES

- All dimensions in millimetres. The datum symbol marks the origin: the board's lower-left corner, X right, Y up, seen from the component side.
- Geometry is design nominal, read from the KiCad board file.
- Hole IDs are the board's own reference designators from the KiCad file, not assigned by this drawing, and are in no positional order.
- The title block reads "TinyTapeout 1,2,3 Demo Board", but no PCB served TT01: that was a bare-die trial run.
- This sheet covers revisions 2.2.5 and 2.2.6: the same board mechanically, compared feature by feature before merging. Only the electrical design and the shuttle differ.
- The Pmod host headers along the lower edge are what a mounting plate registers against. Their 22.86 mm (0.9 in) pitch, per the Digilent Pmod Interface Specification 1.2.0, is the same on every revision; their distance from the lower edge and their position along it are not.
- Port names differ by family: JA, JB, JC on the adapter and Raspberry Pi sheets, by signal direction on the demo boards, PMOD 1 to 3 on the plate sheets. TT-MP-02 maps them.
- Board thickness is 1.6 mm nominal. The KiCad file's 1.60000 mm is its stackup sum, not a specified finished thickness.

NOTES (continued)

- Assembled envelope, connector overhang included, is 104.50 x 82.25 mm against a 104.50 x 81.00 mm board outline. The FEATURE SCHEDULE gives the extents.
- GENERAL TOLERANCE in the title block is a normal board-house figure, not one any source here states.
- Generated from the listed sources and not checked by a second party: there is no CHECKED field because nobody has signed it.

SOURCES

- KiCad board file:
https://github.com/TinyTapeout/tt123-demo-pcb/mpw-mb1.kicad_pcb @ 303509a - title block: TinyTapeout 1,2,3 Demo Board rev 2.2.5, dated 2023-11-13
- KiCad board file:
https://github.com/TinyTapeout/tt123-demo-pcb/mpw-mb1.kicad_pcb @ 3d721a6 - title block: TinyTapeout 1,2,3 Demo Board rev 2.2.6, dated 2024-02-14
- Shuttle mapping: https://docs.google.com/spreadsheets/d/1xD_uemuQcpUnXRY4twOrFzF0urJLVWMDqAnrOJLetM/edit?gid=1250963370 - used by: TT02, TT03

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	3.75	3.75	3.20	6.40
MT2	100.75	3.75	3.20	6.40
MT3	3.75	77.25	3.20	6.40
MT4	100.75	77.25	3.20	6.40

PMOD HOST HEADERS

PORT	REF	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
PMOD A	J3	bottom	41.850	4.465	48.200	5.735
PMOD B	J9	bottom	64.710	4.465	71.060	5.735

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C power / control	8.41 to 18.55	-1.25 to 6.65
2	7-segment display	77.91 to 85.91	24.16 to 36.16
3	LED D1	85.25 to 86.72	76.92 to 79.88
4	LED D2	80.27 to 81.73	77.02 to 79.98
5	LED D3	75.75 to 77.22	76.92 to 79.88
6	LED D4	72.52 to 75.48	70.27 to 71.73
7	8-way input DIP switch	10.80 to 22.20	46.05 to 68.55
8	9-way DIP switch SW2	10.90 to 22.90	17.00 to 43.00
9	Side Pmod, first position - not on this board	-	-
10	Side Pmod, second position - not on this board	-	-
11	Side Pmod, third position - not on this board	-	-

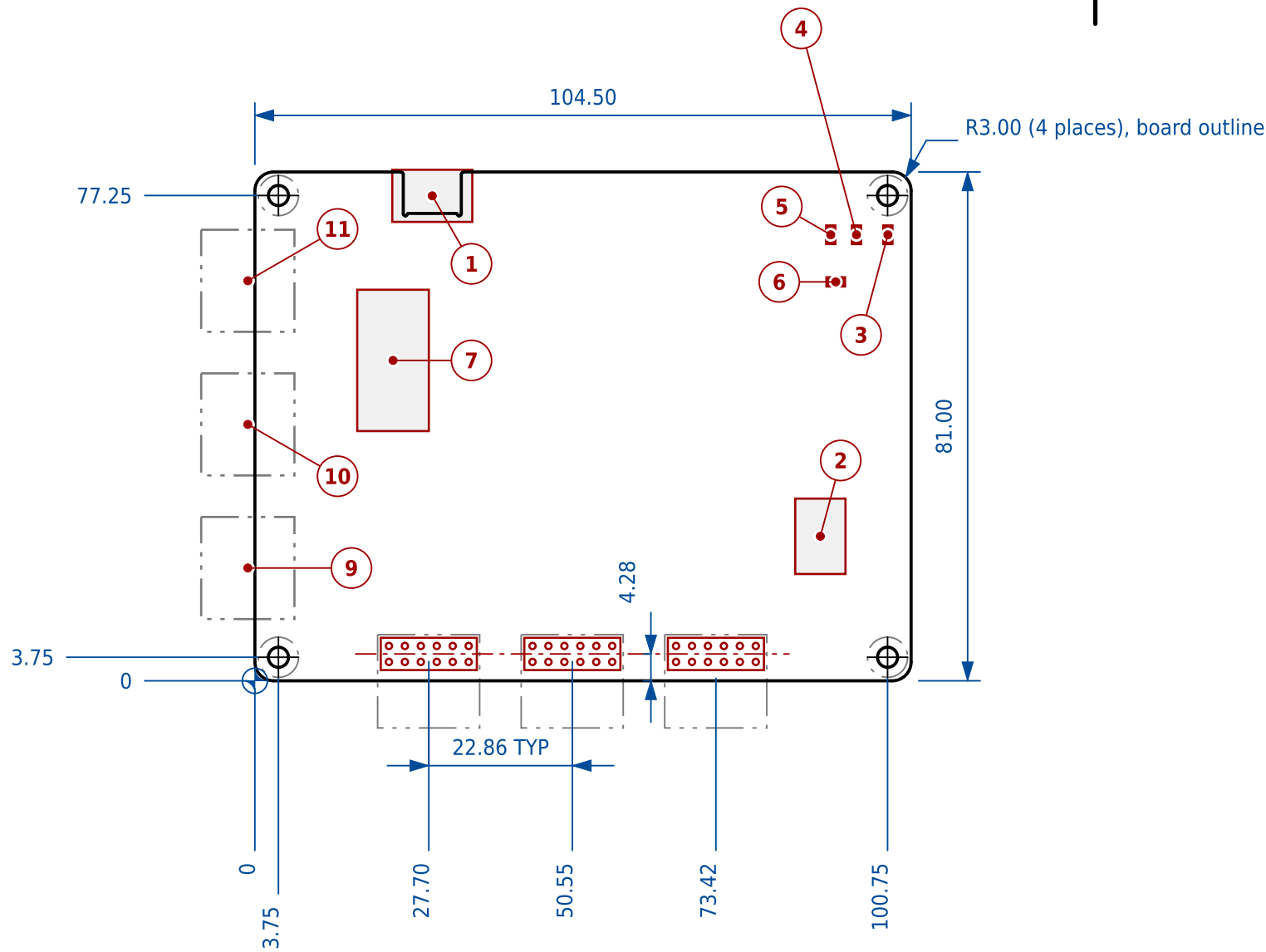
LEGEND

- Board outline
- Component body, scheduled
- Pmod connector body, hole keep-out
- Dimension, extension and leader

DB MPW V2.2.5 / DB MPW V2.2.6

mpw-mb1 rev 2.2.5 and 2.2.6

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-101-g27417b1	mm	1:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	TT-DB-01	A
MATERIAL		GENERAL TOLERANCE	
PCB, 1.6 nominal		edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08	



NOTES

- All dimensions in millimetres. The datum symbol marks the origin: the board's lower-left corner, X right, Y up, seen from the component side.
- Geometry is design nominal, read from the KiCad board file.
- Hole IDs are the board's own reference designators from the KiCad file, not assigned by this drawing, and are in no positional order.
- v1.2.2c is an assembly variant, not a layout change: no X1, a 0R at R51. Its KiCad title block reads rev 1.2.3.
- This sheet covers revisions 1.2.2 and 1.2.3: the same board mechanically, compared feature by feature before merging. Only the electrical design and the shuttle differ.
- The Pmod host headers along the lower edge are what a mounting plate registers against. Their 22.86 mm (0.9 in) pitch, per the Digilent Pmod Interface Specification 1.2.0, is the same on every revision; their distance from the lower edge and their position along it are not.
- Port names differ by family: JA, JB, JC on the adapter and Raspberry Pi sheets, by signal direction on the demo boards, PMOD 1 to 3 on the plate sheets. TT-MP-02 maps them.
- Board thickness is 1.6 mm nominal. The KiCad file's 1.56252 mm is its stackup sum, not a specified finished thickness.

NOTES (continued)

- Assembled envelope, connector overhang included, is 104.50 x 81.35 mm against a 104.50 x 81.00 mm board outline. The FEATURE SCHEDULE gives the extents.
- Upper edge carries a shallow recess for the USB-C shell, contributed by the connector footprint's own edge cuts.
- GENERAL TOLERANCE in the title block is a normal board-house figure, not one any source here states.
- Generated from the listed sources and not checked by a second party: there is no CHECKED field because nobody has signed it.

SOURCES

- KiCad board file: https://github.com/TinyTapeout/tt-demo-pcb/tinytapeout-demo.kicad_pcb @ cfdd80d7b - title block: TinyTapeout 4+ Demoboard rev 1.2.2, dated 2024-04-26
- KiCad board file: https://github.com/TinyTapeout/tt-demo-pcb/tinytapeout-demo.kicad_pcb @ a88cbc08b - title block: TinyTapeout 4+ Demoboard rev 1.2.3, dated 2024-06-11
- Shuttle mapping: https://docs.google.com/spreadsheets/d/1xD_uemuQcpUnXRY4twOrFzF0urJLVWMDqAnrOJLetM/edit?gid=1250963370 - used by: TT04, TT05

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	3.75	3.75	3.20	6.40
MT2	100.75	3.75	3.20	6.40
MT3	3.75	77.25	3.20	6.40
MT4	100.75	77.25	3.20	6.40

PMOD HOST HEADERS

PORT	REF	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
INPUT	J3	bottom	27.695	4.275	34.045	5.545
BIDIR	J5	bottom	50.555	4.275	56.905	5.545
OUTPUT	J6	bottom	73.415	4.275	79.765	5.545

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C power / control	21.88 to 34.62	73.05 to 81.35
2	7-segment display	86.05 to 94.05	17.00 to 29.00
3	LED D1	100.07 to 101.53	69.52 to 72.48
4	LED D2	95.07 to 96.53	69.52 to 72.48
5	LED D3	90.97 to 92.43	69.52 to 72.48
6	LED D4	91.02 to 93.98	62.77 to 64.23
7	8-way input DIP switch	16.30 to 27.70	39.75 to 62.25
8	Second DIP switch - not on this board	-	-
9	Side Pmod J12 (not fitted)	-8.55 to 6.30	9.83 to 26.08
10	Side Pmod J13 (not fitted)	-8.55 to 6.30	32.69 to 48.94
11	Side Pmod J14 (not fitted)	-8.55 to 6.30	55.55 to 71.80

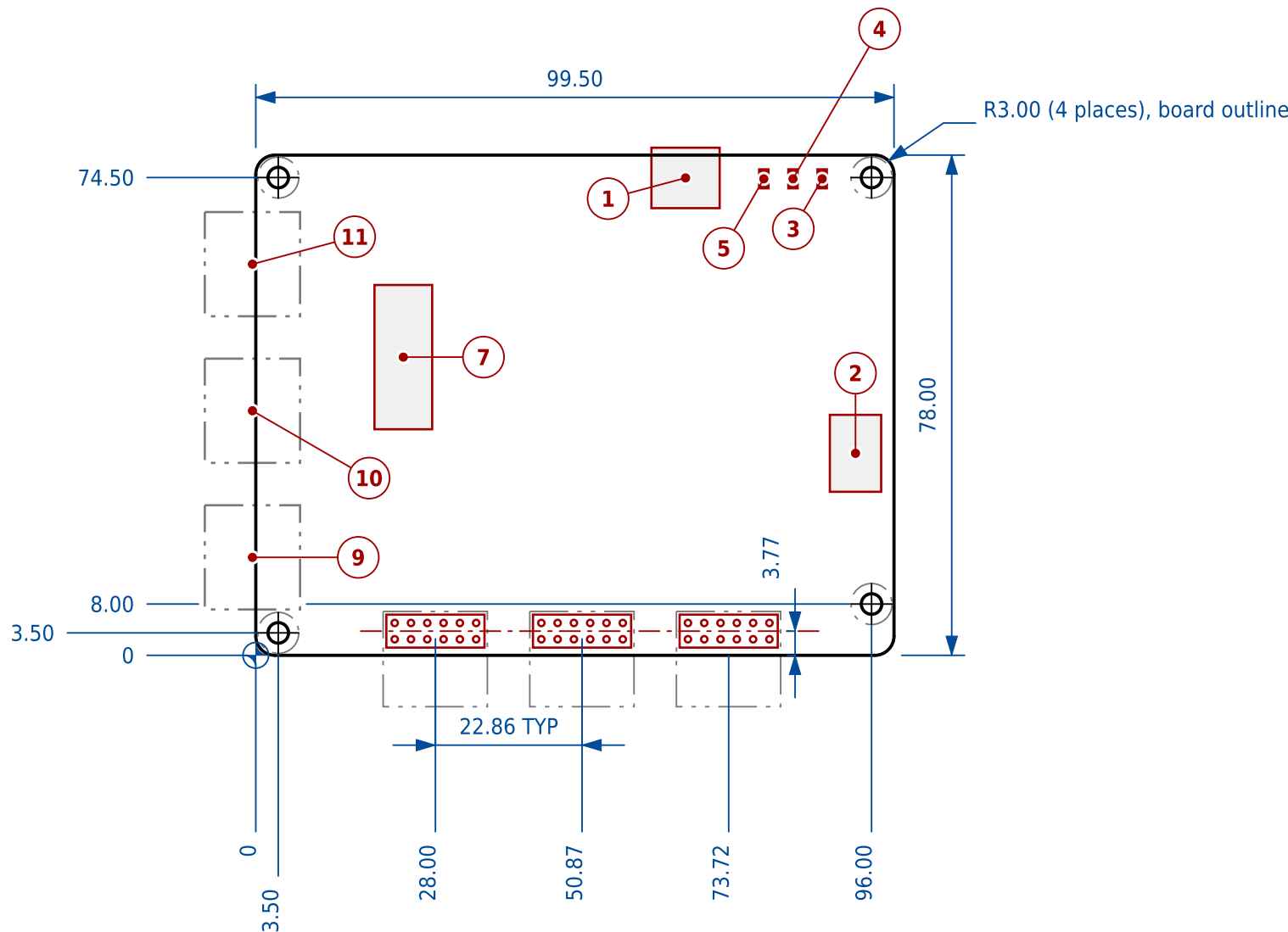
LEGEND

- Board outline
- Component body, scheduled
- Position not fitted on this revision
- Pmod connector body, hole keep-out
- Dimension, extension and leader

DB 4+ V1.2.2 / DB 4+ V1.2.2C

tinytapeout-demo rev 1.2.2 and 1.2.3

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-101-g27417b1	mm	1:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	TT-DB-02	A
MATERIAL		GENERAL TOLERANCE	
PCB, 1.6 nominal		edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08	



NOTES

- All dimensions in millimetres. The datum symbol marks the origin: the board's lower-left corner, X right, Y up, seen from the component side.
- Geometry is design nominal, read from the KiCad board file.
- Hole IDs are the board's own reference designators from the KiCad file, not assigned by this drawing, and are in no positional order.
- This sheet covers revisions 2.0.1 and 2.1.0: the same board mechanically, compared feature by feature before merging. Only the electrical design and the shuttle differ.
- The Pmod host headers along the lower edge are what a mounting plate registers against. Their 22.86 mm (0.9 in) pitch, per the Digilent Pmod Interface Specification 1.2.0, is the same on every revision; their distance from the lower edge and their position along it are not.
- Port names differ by family: JA, JB, JC on the adapter and Raspberry Pi sheets, by signal direction on the demo boards, PMOD 1 to 3 on the plate sheets. TT-MP-02 maps them.
- Board thickness is 1.6 mm nominal. The KiCad file's 1.56252 mm is its stackup sum, not a specified finished thickness.

NOTES (continued)

- Assembled envelope, connector overhang included, is 99.50 x 79.15 mm against a 99.50 x 78.00 mm board outline. The FEATURE SCHEDULE gives the extents.
- GENERAL TOLERANCE in the title block is a normal board-house figure, not one any source here states.
- Generated from the listed sources and not checked by a second party: there is no CHECKED field because nobody has signed it.

SOURCES

- KiCad board file: https://github.com/TinyTapeout/tt-demo-pcb/tinytapeout-demo.kicad_pcb @ 292760e1f - title block: TinyTapeout 06+ Demoboard rev 2.0.1, dated 2024-09-05
- KiCad board file: https://github.com/TinyTapeout/tt-demo-pcb/tinytapeout-demo.kicad_pcb @ a799acb38 - title block: TinyTapeout 06+ Demoboard rev 2.1.0, dated 2024-12-10
- Shuttle mapping: https://docs.google.com/spreadsheets/d/1xD_uemuQcpUnXRY4twOrFzF0urJLVWMDqAnrOJLetM/edit?gid=1250963370 - used by: TT06, TT07

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	3.50	3.50	3.20	6.40
MT2	96.00	8.00	3.20	6.40
MT3	3.50	74.50	3.20	6.40
MT4	96.00	74.50	3.20	6.40

PMOD HOST HEADERS

PORT	REF	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
INPUT	J3	bottom	28.005	3.775	34.355	5.045
BIDIR	J5	bottom	50.865	3.775	57.215	5.045
OUTPUT	J6	bottom	73.725	3.775	80.075	5.045

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C power / control	61.68 to 72.32	69.73 to 79.15
2	7-segment display	89.50 to 97.50	25.50 to 37.50
3	LED D1	87.57 to 89.03	72.82 to 75.78
4	LED D2	83.02 to 84.48	72.82 to 75.78
5	LED D3	78.47 to 79.93	72.82 to 75.78
6	Fourth indicator LED - not on this board	-	-
7	8-way input DIP switch	18.50 to 27.50	35.25 to 57.75
8	Second DIP switch - not on this board	-	-
9	Side Pmod J12 (not fitted)	-7.95 to 6.90	7.15 to 23.40
10	Side Pmod J13 (not fitted)	-7.95 to 6.90	30.01 to 46.26
11	Side Pmod J14 (not fitted)	-7.95 to 6.90	52.87 to 69.12

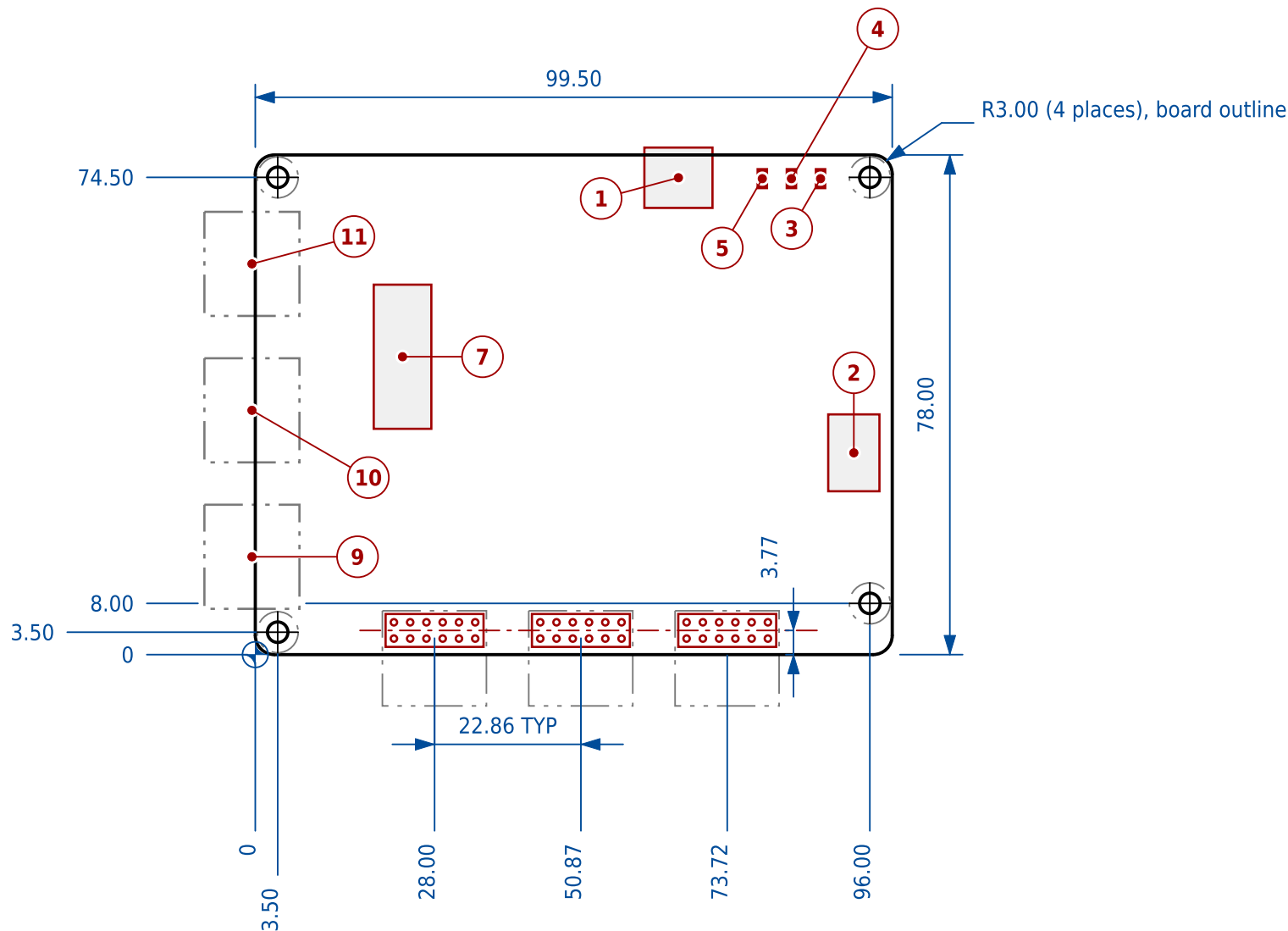
LEGEND

- Board outline
- Component body, scheduled
- Position not fitted on this revision
- Pmod connector body, hole keep-out
- Dimension, extension and leader

DB 06+ V2.0.1 / DB 06+ V2.1.0

tinytapeout-demo rev 2.0.1 and 2.1.0

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-101-g27417b1	mm	1:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	TT-DB-03	A
MATERIAL		GENERAL TOLERANCE	
PCB, 1.6 nominal		edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08	



NOTES

- All dimensions in millimetres. The datum symbol marks the origin: the board's lower-left corner, X right, Y up, seen from the component side.
- Geometry is design nominal, read from the KiCad board file.
- Hole IDs are the board's own reference designators from the KiCad file, not assigned by this drawing, and are in no positional order.
- No other demo board revision shares this geometry.
- The Pmod host headers along the lower edge are what a mounting plate registers against. Their 22.86 mm (0.9 in) pitch, per the Digilent Pmod Interface Specification 1.2.0, is the same on every revision; their distance from the lower edge and their position along it are not.
- Port names differ by family: JA, JB, JC on the adapter and Raspberry Pi sheets, by signal direction on the demo boards, PMOD 1 to 3 on the plate sheets. TT-MP-02 maps them.
- Board thickness is 1.6 mm nominal. The KiCad file's 1.56252 mm is its stackup sum, not a specified finished thickness.
- Assembled envelope, connector overhang included, is 99.50 x 79.15 mm against a 99.50 x 78.00 mm board outline. The FEATURE SCHEDULE gives the extents.

NOTES (continued)

- GENERAL TOLERANCE in the title block is a normal board-house figure, not one any source here states.
- Generated from the listed sources and not checked by a second party: there is no CHECKED field because nobody has signed it.

SOURCES

- KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad_pcb @ 028a51b1e - title block: TinyTapeout 06+ Demoboard rev 2.1.2, dated 2025-02-11
- Shuttle mapping: https://docs.google.com/spreadsheets/d/1xD_uemuQcpUnXRY4twOrFzF0urJLVWMDqAnrOJLetM/edit?gid=1250963370 - row "DB 06+ v2.1.2", Used by: TT08.

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	3.50	3.50	3.20	6.40
MT2	96.00	8.00	3.20	6.40
MT3	3.50	74.50	3.20	6.40
MT4	96.00	74.50	3.20	6.40

PMOD HOST HEADERS

PORT	REF	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
INPUT	J3	bottom	28.005	3.775	34.355	5.045
BIDIR	J5	bottom	50.865	3.775	57.215	5.045
OUTPUT	J6	bottom	73.725	3.775	80.075	5.045

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C power / control	60.78 to 71.42	69.73 to 79.15
2	7-segment display	89.50 to 97.50	25.50 to 37.50
3	LED D1	87.57 to 89.03	72.82 to 75.78
4	LED D2	83.02 to 84.48	72.82 to 75.78
5	LED D3	78.47 to 79.93	72.82 to 75.78
6	Fourth indicator LED - not on this board	-	-
7	8-way input DIP switch	18.50 to 27.50	35.25 to 57.75
8	Second DIP switch - not on this board	-	-
9	Side Pmod J12 (not fitted)	-7.95 to 6.90	7.15 to 23.40
10	Side Pmod J13 (not fitted)	-7.95 to 6.90	30.01 to 46.26
11	Side Pmod J14 (not fitted)	-7.95 to 6.90	52.87 to 69.12

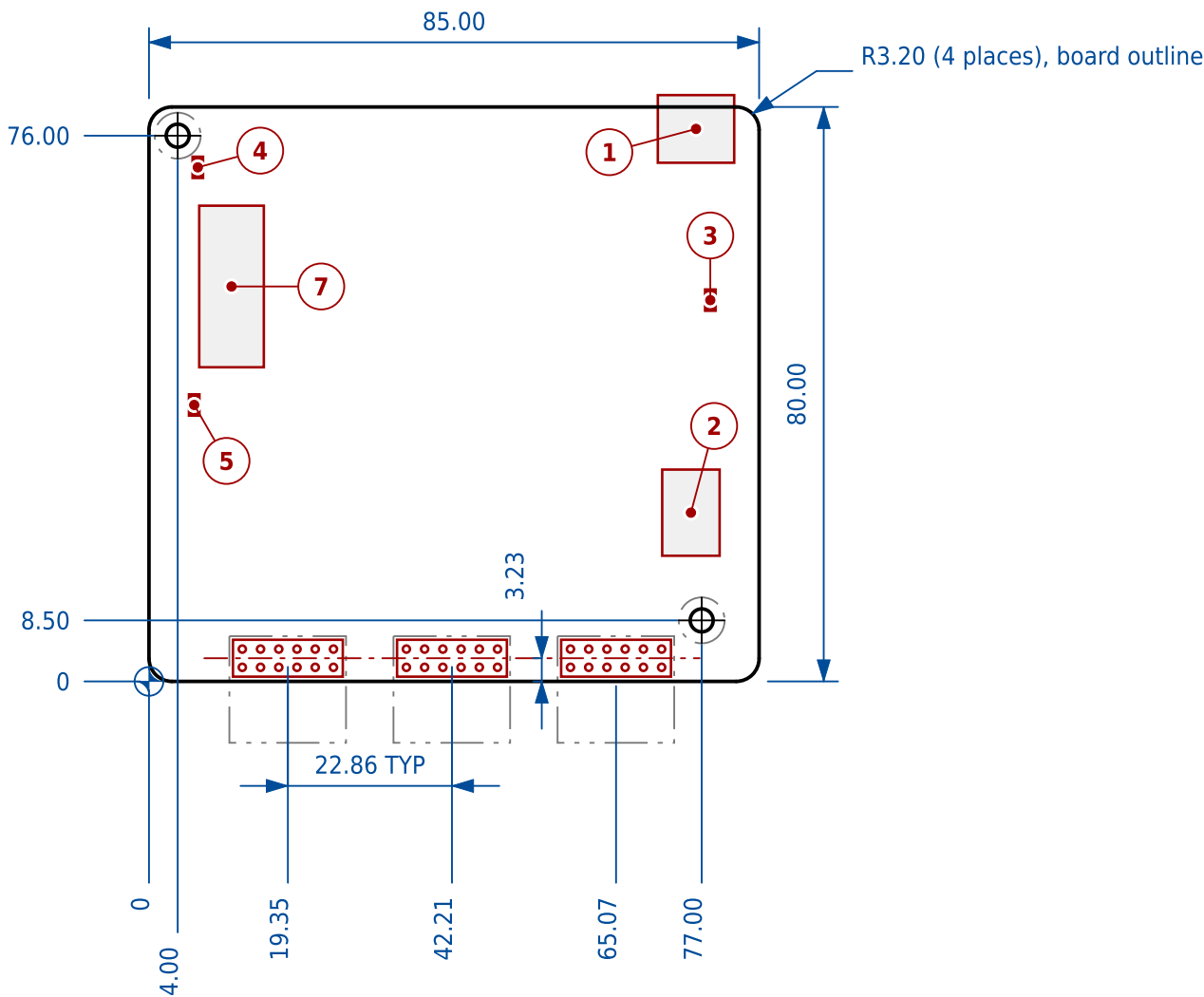
LEGEND

- Board outline
- Component body, scheduled
- Position not fitted on this revision
- Pmod connector body, hole keep-out
- Dimension, extension and leader

DB 06+ V2.1.2

tinytapeout-demo rev 2.1.2

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-101-g27417b1	mm	1:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	TT-DB-04	A
MATERIAL		GENERAL TOLERANCE	
PCB, 1.6 nominal		edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08	



NOTES

- All dimensions in millimetres. The datum symbol marks the origin: the board's lower-left corner, X right, Y up, seen from the component side.
- Geometry is design nominal, read from the KiCad board file.
- Hole IDs are the board's own reference designators from the KiCad file, not assigned by this drawing, and are in no positional order.
- No other demo board revision shares this geometry.
- ETR is Tiny Tapeout's own designation for this board. The KiCad title block reads "Tiny Tapeout Demoboard v3" and does not carry it; the name here is the board's ID in Tiny Tapeout's board revision spreadsheet. Nothing mechanical depends on it.
- The Pmod host headers along the lower edge are what a mounting plate registers against. Their 22.86 mm (0.9 in) pitch, per the Digilent Pmod Interface Specification 1.2.0, is the same on every revision; their distance from the lower edge and their position along it are not.
- Port names differ by family: JA, JB, JC on the adapter and Raspberry Pi sheets, by signal direction on the demo boards, PMOD 1 to 3 on the plate sheets. TT-MP-02 maps them.

NOTES (continued)

- Board thickness is 1.6 mm nominal. The KiCad file's 1.56100 mm is its stackup sum, not a specified finished thickness.
- Assembled envelope, connector overhang included, is 85.00 x 81.65 mm against a 85.00 x 80.00 mm board outline. The FEATURE SCHEDULE gives the extents.
- GENERAL TOLERANCE in the title block is a normal board-house figure, not one any source here states.
- Generated from the listed sources and not checked by a second party: there is no CHECKED field because nobody has signed it.

SOURCES

- KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad_pcb @ d830790ca - title block: Tiny Tapeout Demoboard v3 rev 3.2, dated 2025-11-12
- Shuttle mapping: https://docs.google.com/spreadsheets/d/1xD_uemuQcpUnXRY4twOrFzF0urJLVWMDqAnrOJLetM/edit?gid=1250963370 - row "DB ETR v3.2", Used by: TT09, TTSKY25a, TTSKY25b, TTGF0p2.

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	4.00	76.00	3.20	6.40
MT2	77.00	8.50	3.20	6.40

PMOD HOST HEADERS

PORT	REF	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
INPUT	J11	bottom	19.350	3.230	25.700	4.500
BIDIR	J12	bottom	42.210	3.230	48.560	4.500
OUTPUT	J13	bottom	65.070	3.230	71.420	4.500

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C power / control	70.89 to 81.53	72.23 to 81.65
2	7-segment display	71.50 to 79.50	17.50 to 29.50
3	LED D3	77.47 to 78.93	51.62 to 54.58
4	LED D4	6.07 to 7.53	70.11 to 73.07
5	LED D5	5.57 to 7.03	37.02 to 39.98
6	Fourth indicator LED - not on this board	-	-
7	8-way input DIP switch	7.00 to 16.00	43.75 to 66.25
8	Second DIP switch - not on this board	-	-
9	Side Pmod, first position - not on this board	-	-
10	Side Pmod, second position - not on this board	-	-
11	Side Pmod, third position - not on this board	-	-

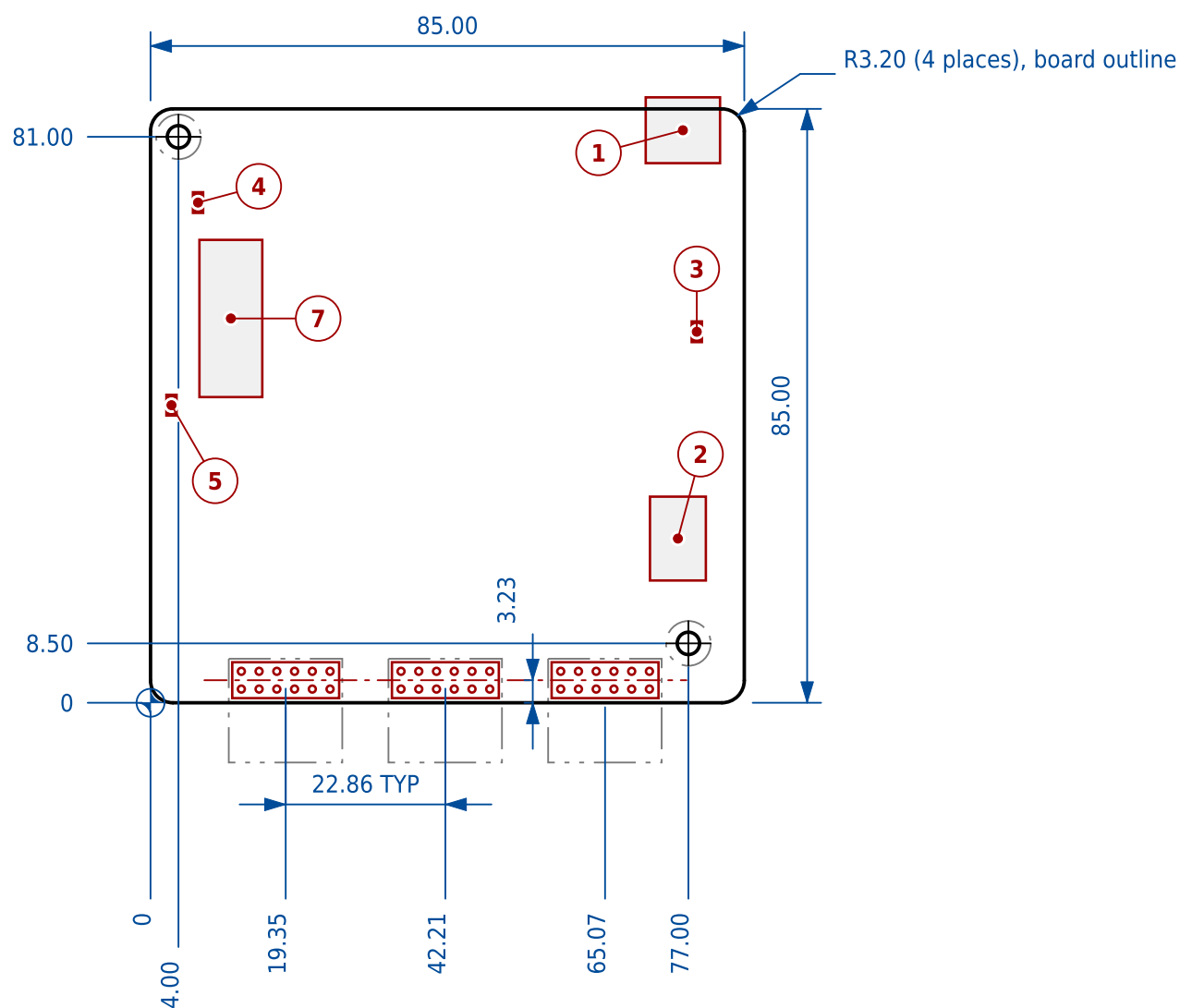
LEGEND

- Board outline
- Component body, scheduled
- Pmod connector body, hole keep-out
- Dimension, extension and leader

DB ETR V3.2

tinytapeout-demo rev 3.2

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-101-g27417b1	mm	1:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	TT-DB-05	A
MATERIAL		GENERAL TOLERANCE	
PCB, 1.6 nominal		edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08	



NOTES

- All dimensions in millimetres. The datum symbol marks the origin: the board's lower-left corner, X right, Y up, seen from the component side.
- Geometry is design nominal, read from the KiCad board file.
- Hole IDs are the board's own reference designators from the KiCad file, not assigned by this drawing, and are in no positional order.
- No other demo board revision shares this geometry.
- This revision has no row in Tiny Tapeout's board revision spreadsheet, which lists DB ETR v3.2 as the current demoboard. The name used here follows that ID; no shuttle is known to have shipped on it.
- The Pmod host headers along the lower edge are what a mounting plate registers against. Their 22.86 mm (0.9 in) pitch, per the Digilent Pmod Interface Specification 1.2.0, is the same on every revision; their distance from the lower edge and their position along it are not.
- Port names differ by family: JA, JB, JC on the adapter and Raspberry Pi sheets, by signal direction on the demo boards, PMOD 1 to 3 on the plate sheets. TT-MP-02 maps them.
- Board thickness is 1.6 mm nominal. The KiCad file's 1.56100 mm is its stackup sum, not a specified finished thickness.

NOTES (continued)

- Assembled envelope, connector overhang included, is 85.00 x 86.65 mm against a 85.00 x 85.00 mm board outline. The FEATURE SCHEDULE gives the extents.
- GENERAL TOLERANCE in the title block is a normal board-house figure, not one any source here states.
- Generated from the listed sources and not checked by a second party: there is no CHECKED field because nobody has signed it.

SOURCES

- KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad_pcb @ ecb636ace - title block: Tiny Tapeout Demoboard v3 rev 3.3, dated 2026-06-23
- Shuttle mapping: https://docs.google.com/spreadsheets/d/1xD_uemuQcpUnXRY4twOrFzF0urJLVWMdqAnrOJLetM/edit?gid=1250963370 - no row: this revision is not in the board revision spreadsheet, so no shuttle is recorded for it.

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	4.00	81.00	3.20	6.40
MT2	77.00	8.50	3.20	6.40

PMOD HOST HEADERS

PORT	REF	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
INPUT	J11	bottom	19.350	3.230	25.700	4.500
BIDIR	J12	bottom	42.210	3.230	48.560	4.500
OUTPUT	J13	bottom	65.070	3.230	71.420	4.500

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C power / control	70.89 to 81.53	77.23 to 86.65
2	7-segment display	71.50 to 79.50	17.50 to 29.50
3	LED D3	77.47 to 78.93	51.62 to 54.58
4	LED D4	6.07 to 7.53	70.11 to 73.07
5	LED D5	2.27 to 3.73	41.11 to 44.07
6	Fourth indicator LED - not on this board	-	-
7	8-way input DIP switch	7.00 to 16.00	43.75 to 66.25
8	Second DIP switch - not on this board	-	-
9	Side Pmod, first position - not on this board	-	-
10	Side Pmod, second position - not on this board	-	-
11	Side Pmod, third position - not on this board	-	-

LEGEND

- Board outline
- Component body, scheduled
- Pmod connector body, hole keep-out
- Dimension, extension and leader

DB ETR V3.3

tinytapeout-demo rev 3.3

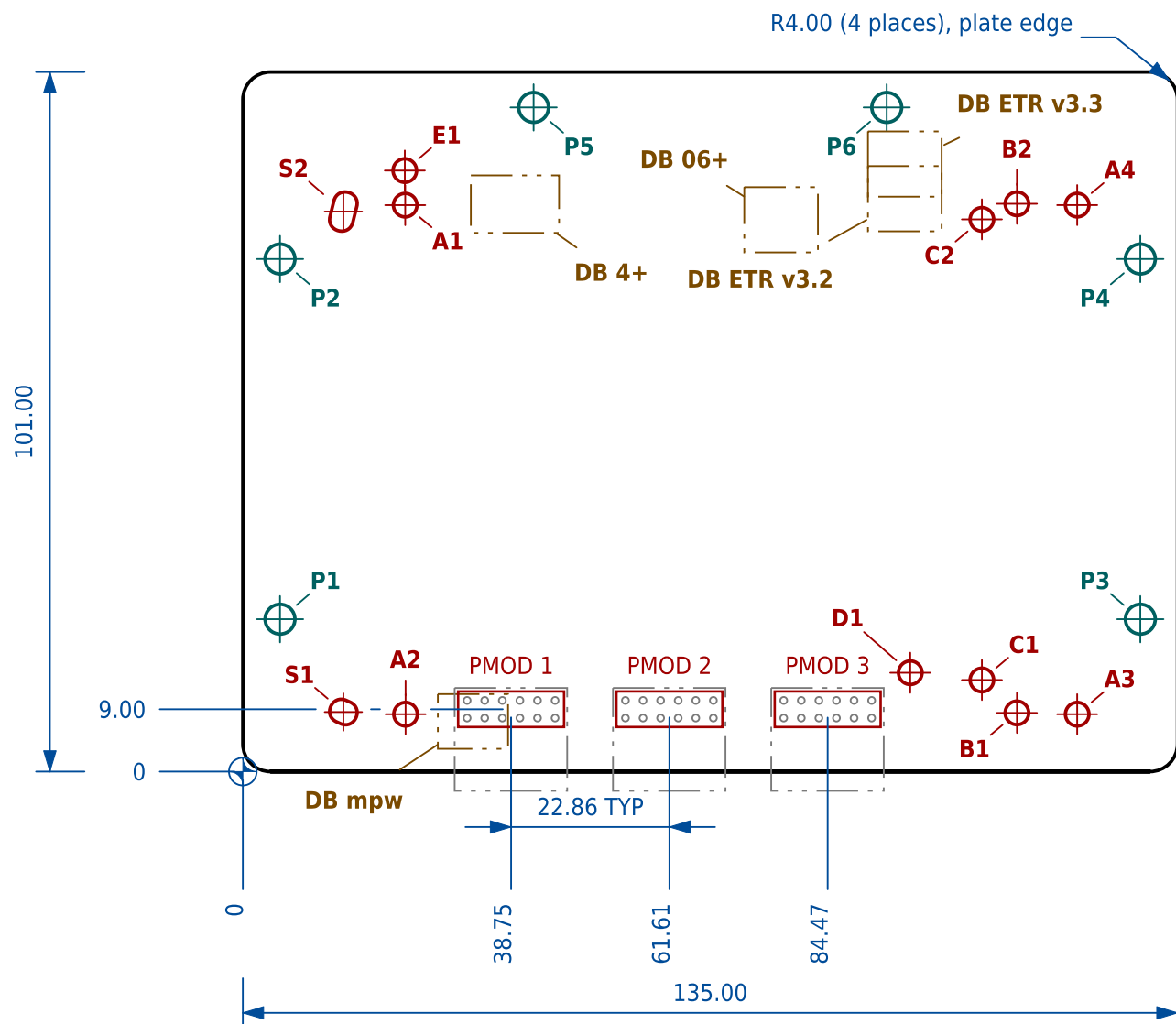
DRAWN	VERSION	UNITS	SCALE
generated	v0.0-101-g27417b1	mm	1:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	TT-DB-06	A
MATERIAL		GENERAL TOLERANCE	
PCB, 1.6 nominal		edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08	

A

B

C

D



NOTES

- All dimensions in millimetres. The datum symbol marks the origin: the plate's lower-left corner, X right, Y up, seen from the side the board mounts on.
- Hole positions are tabulated, not dimensioned on the view: there are too many to dimension and keep it readable. A slot row gives its two end centres; DIA/WIDTH is the slot width, LENGTH is overall.
- USED BY reads <group>:<hole>; MT1 to MT4 are the board's own hole IDs. Groups, by board revision: DB mpw = tt123-v2.2.5, tt123-v2.2.6; DB 4+ = v1.2.2, v1.2.3; DB 06+ = v2.0.1, v2.1.0, v2.1.2; DB ETR v3.2 = v3.2; DB ETR v3.3 = themselves, no shuttle yet.
- A board hole's ID letter is the revision it is there for -- A DB mpw, B DB 4+, C DB 06+, D DB ETR v3.2, E DB ETR v3.3 -- numbered within it. S is a slot and P a plate fixing. Four features serve two revisions; each keeps one ID.
- The three PMOD envelopes are not machined features. They mark where the Pmod host pin fields end up, the same place for every board revision. That is the point of the plate.
- Fit the board on standoffs. A board uses only the holes and slots its USED BY row names.

NOTES (continued)

- Board holes and slots are 3.4 mm, a close clearance fit on M3. Plate fixings are 4.3 mm, a clearance fit on M4. A slot spans two revisions whose holes are too close together to drill separately.
- The Pmod host connector bodies overhang the plate's front edge by 2.78 mm, so a peripheral module plugs into clear air.
- Cut file: tt-generic-mounting-plate.dxf, four layers. Sizes here are finished sizes; allow for your cutter's kerf.
- Fit the plate to its chassis before the board: the fixings sit in the border, which a board overhangs. Least material between features is 1.59 mm, between A1 and E1.
- Drawing TT-MP-02 shows which holes each revision uses.

SOURCES

- Derived from: tinytapeout/boards.py - Board outlines, mounting holes and Pmod host positions for every Tiny Tapeout demo board revision, extracted from the upstream KiCad files.

BOARD MOUNTING HOLES

ID	X mm	Y mm	DIA/WIDTH mm	LENGTH mm	USED BY
A1	23.45	81.78	3.40	-	DB mpw:MT3, DB ETR v3.2:MT1
A2	23.51	8.29	3.40	-	DB mpw:MT1
A3	120.51	8.29	3.40	-	DB mpw:MT2
A4	120.51	81.78	3.40	-	DB mpw:MT4
B1	111.81	8.47	3.40	-	DB 4+:MT2
B2	111.81	81.97	3.40	-	DB 4+:MT4
C1	106.75	13.22	3.40	-	DB 06+:MT2
C2	106.75	79.72	3.40	-	DB 06+:MT4
D1	96.40	14.27	3.40	-	DB ETR v3.2:MT2, DB ETR v3.3:MT2
E1	23.40	86.77	3.40	-	DB ETR v3.3:MT1
S1	14.24 / 14.80	8.72 / 8.47	3.40	4.01	DB 4+:MT1, DB 06+:MT1
S2	14.24 / 14.80	79.72 / 81.97	3.40	5.72	DB 4+:MT3, DB 06+:MT3

PLATE FIXING HOLES

ID	X mm	Y mm	DIA mm
P1	5.37	22.00	4.30
P2	5.37	74.00	4.30
P3	129.63	22.00	4.30
P4	129.63	74.00	4.30
P5	42.00	95.89	4.30
P6	93.00	95.89	4.30

LEGEND

- Plate edge
- Hole or slot for a demo board fastener
- Plate fixing, into the chassis
- Slot axis
- Pmod host pin field, where a board's hosts land
- Pmod connector body, overhanging the front edge
- USB-C connector, one position per board version
- Dimension, extension and leader

SOURCES (continued)

- Pmod host pitch:
https://digilent.com/reference/_media/reference/pmod/pmod-interface-specification-1_2_0.pdf - Digilent mandate .90 in (22.86 mm) between adjacent host ports on a board edge, which is why one plate can serve every revision.

TT GENERIC MOUNTING PLATE

Accepts every demo board revision, Pmod hosts fixed in place

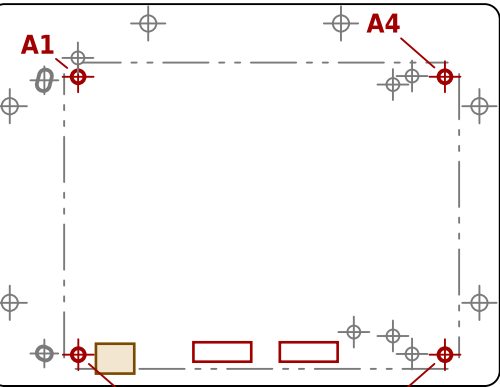
DRAWN	VERSION	UNITS	SCALE
generated	v0.0-101-g27417b1	mm	1:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 2	TT-MP-01	A
MATERIAL		GENERAL TOLERANCE	
3 mm acrylic or 1.6 mm FR4		edge +/-0.20 hole pos +/-0.10 allow for cutter kerf	

A

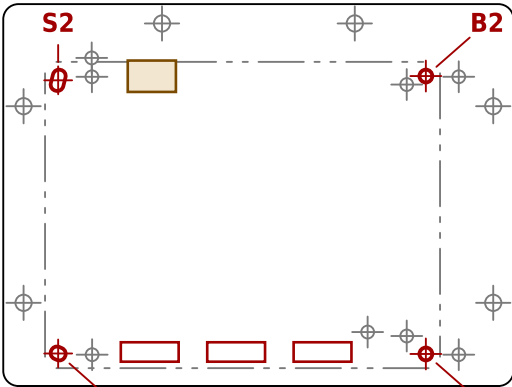
B

C

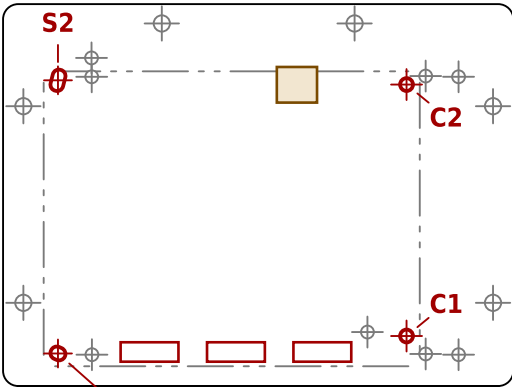
D



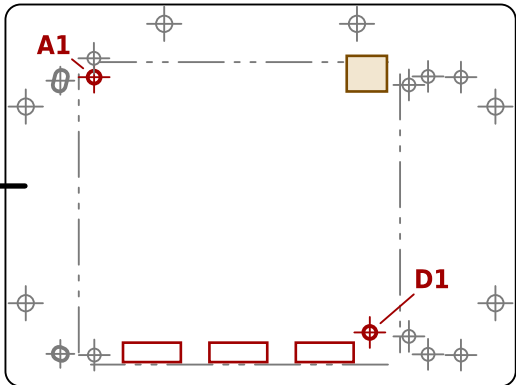
DB mpw
TT02, TT03
board rev tt123-v2.2.5, tt123-v2.2.6



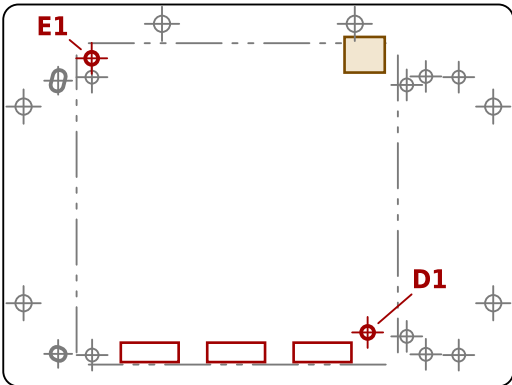
DB 4+
TT04, TT05
board rev v1.2.2, v1.2.3



DB 06+
TT06, TT07, TT08
board rev v2.0.1, v2.1.0, v2.1.2



DB ETR v3.2
TT09, TTSKY25a, TTSKY25b, TTGF0p2
board rev v3.2



DB ETR v3.3
no shipped shuttle yet
board rev v3.3

NOTES

- Each view shows one group of demo board revisions on the plate, with the holes and slots that group uses drawn solid and labelled, and the rest greyed back. Hole IDs match drawing TT-MP-01.
- The revisions in a group share their mounting holes and Pmod host positions exactly, which is all the plate registers against. They may differ elsewhere: v2.1.2 moved its USB-C connector 0.9 mm relative to v2.0.1 and v2.1.0, for instance.
- dX and dY place the board: add them to a coordinate in that board's own frame to get a plate coordinate.
- The DB mpw board has only two Pmod hosts. They go on plate Pmod positions 2 and 3, which makes the plate 5.15 mm narrower than putting them on 1 and 2 would.
- Every revision puts its Pmod host pin fields on the same three positions, at the same height above the plate's front edge. See drawing TT-MP-01 for the plate itself.

BOARD PLACEMENT ON THE PLATE

BOARD	SHUTTLES	dX	dY	PMODS	HOLES USED
DB mpw	TT02, TT03	19.76	4.54	2, 3	A1, A2, A3, A4
DB 4+	TT04, TT05	11.05	4.72	1, 2, 3	B1, B2, S1, S2
DB 06+	TT06, TT07, TT08	10.74	5.22	1, 2, 3	C1, C2, S1, S2
DB ETR v3.2	TT09, TTSKY25a, TTSKY25b, TTGF0p2	19.40	5.77	1, 2, 3	A1, D1
DB ETR v3.3	not yet shipped	19.40	5.77	1, 2, 3	D1, E1

USB-C CONNECTOR POSITION ON THE PLATE

BOARD	X EXTENT mm	Y EXTENT mm
DB mpw	28.17 to 38.31	3.29 to 11.19
DB 4+	32.94 to 45.67	77.78 to 86.08
DB 06+	72.42 to 83.06	74.95 to 84.38
DB ETR v3.2	90.29 to 100.93	78.00 to 87.42
DB ETR v3.3	90.29 to 100.93	83.00 to 92.42

LEGEND

- Plate edge
- Hole or slot this revision uses
- Holes it does not use, and its board outline
- USB-C connector on this revision

TT MOUNTING PLATE FITTING GUIDE

Which holes each demo board revision uses

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-101-g27417b1	mm	1:2
SIZE	SHEET	DRAWING NO	REV
A3	2 OF 2	TT-MP-02	A
MATERIAL		GENERAL TOLERANCE	
- not a made part		reference only - TT-MP-01 governs every dimension	